

## CS722/822 Machine Learning Fall 2025

### Instructor

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Class times: 3 credit hours. T: 06:00 pm-08:40 pm  
Office Hours: T: 05:00 pm-06:00 pm or by appointment  
Class location:

### Teaching Assistant

### Prerequisites

Linear algebra, basic probability theory, basic data structures and algorithms, computer programming (Python)

### Textbook and References

- Primary Course Material: **Lecture slides and instructor-prepared notes.**
- Textbook (assumed):  
*Aurélien Géron*, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd Edition), O'Reilly Media, 2022. ISBN: 9781098122478

### Optional References (for depth and additional reading)

- *Christopher M. Bishop*, **Pattern Recognition and Machine Learning**, Springer, 2006. ISBN: 9780387310732 (*For mathematical foundations and probabilistic modeling.*)
- *Kevin P. Murphy*, **Machine Learning: A Probabilistic Perspective**, MIT Press, 2012. ISBN: 9780262018029  
(*Advanced theoretical treatment with a focus on probabilistic approaches.*)

### Brief Course Description

This graduate course is a rigorous, hands-on introduction to **classical machine learning**. We cover supervised learning (linear & polynomial regression; logistic/softmax classification; k-NN; LDA/QDA; **SVMs**; **decision trees**; **ensemble methods**) with an emphasis on evaluation, cross-validation, calibration, and error analysis. Unsupervised learning includes **clustering** (k-means, hierarchical, DBSCAN, Gaussian mixtures/anomaly detection) and **dimensionality reduction** (PCA, randomized/incremental PCA, LLE, random projections). **Neural networks are introduced only as perceptron/MLP classifiers** to place them within the ML toolbox. A late-semester **survey** highlights modern directions (deep learning, self-/semi-supervised learning, reinforcement learning, federated/privacy-preserving learning) and points students to dedicated courses. The course includes one curated paper-review session (after instructor approval), a midterm, a team project with a conference-style presentation, and a comprehensive final exam.

### Late Submission Policy

Homework assignments submitted after the deadline will incur a 10% grade deduction per late day. Submissions more than 5 days late will receive a grade of zero, unless prior arrangements have been approved by the instructor.

### Course Learning Objectives

Upon successful completion of this course, students will be able to:

1. **Explain** fundamental ML concepts, problem types, common pitfalls (e.g., leakage), and the end-to-end pipeline from data prep to evaluation and deployment.
2. **Implement and evaluate** classical supervised models (linear/polynomial regression; logistic/softmax; k-NN; LDA/QDA; SVMs; decision trees; ensembles) using sound metrics and scikit-learn pipelines.
3. **Perform model selection** with cross-validation, handle class imbalance, conduct error analysis, and apply basic probability calibration.
4. **Apply unsupervised methods**, including clustering (k-means, hierarchical, DBSCAN, Gaussian mixtures) and dimensionality reduction (PCA, randomized/incremental PCA, LLE, random projections), and **assess** results appropriately.
5. **Explain ANN fundamentals** and **implement a simple MLP classifier**, selecting appropriate activations/losses and basic regularization/early stopping.
6. **Design reproducible ML experiments**, including stratified splits, hyperparameter search, ablations, and documentation/governance practices.
7. **Read and critique** a contemporary ML paper and **present** findings succinctly.
8. **Plan and execute** a team ML project: define the problem, build baselines and improved models, evaluate rigorously, and communicate results.
9. **Describe at a high level** modern directions beyond classical ML (deep learning, self-/semi-supervised learning, reinforcement learning, federated/privacy-preserving learning) and identify when specialized courses/tools are appropriate.

### Grading Policy

| Item                 | Grade Allocated |
|----------------------|-----------------|
| Midterm              | 20              |
| Final Exam           | 30              |
| Homework Assignments | 20              |
| Term Project*        | 30              |

\*The project grade entails the grade of paper and proposal presentation, project presentation, and project report.

### Weekly Topics

| Week              | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Materials          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 1<br>Aug 26, 2025 | <b>Introduction:</b> What is ML?, The Big Picture, Why use ML?, Why We Still Learn Classical ML in 2025, Applications, Types of ML, Main Challenges to ML, Testing and Validating, The end-to-end ML pipeline: (Data Mining, Data Visualization, Data Preparation, Model Selection, Fine-tuning, Launching, and Maintenance),                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>Géron 1, 2</i>  |
| 2<br>Sep 2, 2025  | <b>Preliminaries:</b> Review of linear algebra, probability theory, and optimization theory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |
| 3<br>Sep 9, 2025  | <b>Regression:</b> Linear Regression, Gradient Descent, Polynomial Regression, Learning Curves, Overfitting & Generalization, Regularization Techniques (Ridge, Lasso, Elastic Net, Early Stopping)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Géron 3</i>     |
| 4<br>Sep 16, 2025 | <b>Classification I: Foundational and Classical Methods</b><br>MNIST dataset, <i>overview of classification tasks</i> , training a binary classifier, logistic regression (estimating probabilities, training and cost function, decision boundaries), <b>connection to linear regression as a special case</b> , performance measures (accuracy, precision/recall, F1, ROC/AUC), error analysis (confusion matrix, false positives/negatives), <i>evaluation pipeline (stratified train/validation/test split, cross-validation, leakage check)</i> , KNN (intuition, distance metrics, feature scaling), linear discriminant analysis (generative vs discriminative view), <i>limitations of linear models (non-linear boundaries, XOR example)</i> , multiclass classification (one-vs-rest, one-vs-one), softmax regression, multilabel and multioutput classification, logistic regression – in depth (interpretability, odds ratios) | <i>Géron 3, 10</i> |
| 5<br>Sep 23, 2025 | <b>Classification II: Neural Network–Based Methods</b><br>motivation (limits of linear models, XOR revisited), introduction to artificial neural networks (from biological to artificial neurons, logical computations with neurons), perceptron (capabilities and limits), multilayer perceptron (hidden layers, non-linear decision boundaries), <b>universal approximation perspective (intuition)</b> , backpropagation (high-level intuition, gradient flow), activation functions (sigmoid, tanh, ReLU, softmax), loss functions (log-loss for binary, cross-entropy for multiclass, <b>MSE vs CE discussion</b> ), <b>regularization in neural nets (weight decay, dropout – overview)</b> , wrap-up (neural networks as generalization of logistic regression and foundation for deep learning)                                                                                                                                    |                    |
| 6<br>Sep 30, 2025 | <b>Unsupervised Learning I – Clustering I :</b> (overview, hierarchical clustering, clustering evaluation, silhouette score, spectral clustering), Clustering Algorithms: k-means and DBSCAN: (k-means, k-means++ initialization, Limits of k-means, Using Clustering for Image Segmentation, Using Clustering for Semi-Supervised Learning, feature scaling/standardization, DBSCAN,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Géron 9</i>     |

| Week               | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Materials      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|                    | DBSCAN parameter selection (eps, min_samples), Other Clustering Algorithms),                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |
| 7<br>Oct 7, 2025   | <b>Unsupervised Learning II – Clustering II, Data reduction, and outlier detection:</b> Gaussian Mixtures: (Using Gaussian Mixtures for Anomaly Detection, Selecting the Number of Clusters, Bayesian Gaussian Mixture Models), Anomaly and Novelty Detection, Data representation and compression (Intro)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |
| 8<br>Oct 14, 2025  | NO CLASS (Fall Holiday Sat–Tue)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |
| 9<br>Oct 21, 2025  | <b>Project Proposal and Base Paper Presentation:</b> student teams present a paper of their choice after instructor approval, along with project proposal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |
| 10<br>Oct 28, 2025 | <b>Midterm Exam (60 minutes)</b><br><b>Decision Trees:</b> Training and Visualizing a Decision Tree, Making Predictions, Estimating Class Probabilities, The CART Training Algorithm, Computational Complexity, Gini Impurity or Entropy? Regularization Hyperparameters, Regression, Sensitivity to Axis Orientation, Decision Trees Have a High Variance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Géron 6</i> |
| 11<br>Nov 4, 2025  | NO CLASS (Election Day)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |
| 12<br>Nov 11, 2025 | <b>Support Vector Machines (SVMs):</b> Linear SVM Classification, soft-margin & hinge loss (C, slack variables), Under the Hood of Linear SVM Classifiers, The Dual Problem, Kernelized SVMs, Nonlinear SVM Classification, multiclass strategies (one-vs-rest, one-vs-one), feature scaling, SVM Regression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>Géron 8</i> |
| 13<br>Nov 18, 2025 | <b>Ensemble Learning and Random Forests:</b> (Voting Classifiers, Bagging and Pasting: (Bagging and Pasting in Scikit-Learn, Out-of-Bag Evaluation, Random Patches and Random Subspaces), Random Forests: (Extra-Trees, Feature Importance), Boosting: (AdaBoost, Gradient Boosting, Histogram-Based Gradient Boosting), Stacking)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>Géron 7</i> |
| 14<br>Nov 25, 2025 | <b>Dimensionality Reduction:</b> (The Curse of Dimensionality, Main Approaches for Dimensionality Reduction: (Projection, Manifold Learning), PCA: (Preserving the Variance, Principal Components, Projecting Down to d Dimensions, Using Scikit-Learn, Explained Variance Ratio, Choosing the Right Number of Dimensions, PCA for Compression, Randomized PCA, Incremental PCA), Random Projection, Kernel PCA / LLE / t-SNE, Other Dimensionality Reduction Techniques) Dictionary Learning, Autoencoders<br><b>Beyond ML's Supervised and Unsupervised Learning:</b> (Deep Learning: (DL Concepts and Architectures, The Vanishing/Exploding Gradients Problems, Faster Optimizers, Learning Rate Scheduling, Avoiding Overfitting Through Regularization, Reusing Pretrained Layers), Self-Supervised Learning, Semi-Supervised Learning, Reinforcement Learning, Other ML Learning Paradigms (Incorporating domain knowledge, |                |

| Week              | Topic                                                                                                                                                                                                                                                                           | Materials |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|                   | adaptability, structure, and privacy): (Physics-Informed Machine Learning (PIML), Logic- and Constraint-Guided Learning, Meta-Learning (Learning to Learn), Multi-Modal / Cross-Modal Learning, Process-Centric Learning Paradigms, Federated and Privacy-Preserving Learning)) |           |
| 15<br>Dec 2, 2025 | <b>Project presentations:</b> Student teams present their final term projects                                                                                                                                                                                                   |           |
| Final Week        | A final comprehensive exam                                                                                                                                                                                                                                                      |           |

### Student Mental Health and Wellbeing

ODU is committed to supporting your mental health. The Office of Counseling Services offers free, confidential support including virtual and in-person counseling, group sessions, and crisis services. Same-day or next-day appointments can be scheduled at [odu.edu/counseling-services](https://odu.edu/counseling-services). In case of a mental health emergency—such as thoughts of self-harm, harming others, or recent assault—call 911, or contact a crisis counsellor 24/7 at 757-683-4401 (press Option 2). You may also call the Suicide & Crisis Lifeline at 988 or text "HOME" to 741741.

### Academic Integrity

Old Dominion University expects honesty in all academic work. In this course, your work and conduct must comply with the **Code of Student Conduct** ([odu.edu/oscai](https://odu.edu/oscai)). Academic dishonesty includes:

- **Cheating:** using unauthorized assistance, materials, study aids, or information.
- **Plagiarism:** using others' language or ideas without proper acknowledgment.
- **Fabrication:** inventing, altering, or falsifying data, citations, or information.
- **Facilitation:** helping another student commit a violation or failing to report suspected violations.

Classroom disruptions that undermine instruction are also prohibited. **Suspected violations will be reported to the Office of Student Conduct & Academic Integrity and may result in sanctions up to and including expulsion.**

### Honor Code

Students must follow the ODU Honor Code for all assignments and exams. Collaboration on ideas is encouraged, but **all submitted work (code, text, analysis) must be your own**. Limited use of tools (e.g., ChatGPT) for brainstorming is acceptable; **do not submit AI-generated text/code as your own**. Suspected violations will be handled under University policy.

### Drop Policy

As per University guidelines. See the University Calendar for drop dates.

### Accessibility & Accommodations

ODU provides reasonable accommodations under the ADA. If you need accommodations, **obtain an accommodation letter from the Office of Educational Accessibility (OEA)** and share it with me early so we can implement the approved arrangements. If you anticipate barriers but don't yet have a letter, **contact OEA** to discuss eligibility.

- **OEA:** 1021 Student Success Center, (757) 683-4655, [odu.edu/educationalaccessibility](https://odu.edu/educationalaccessibility)
- Accommodations begin **once** I receive your official OEA letter.

**Attendance Policy**

Students are expected to attend classes regularly.